

Problem M

Backpack Backpack

Time limit: 3 seconds

Memory limit: 1024 megabytes

Problem Description

Today is a wonderful day! Dorara is going out to play, and she wants to bring her favorite toys with her. However, she has so many toys that she is having a hard time deciding which ones to take.

Her older brother, Diegogo, sees her struggling and gives her a piece of advice:
“Since you can’t decide, why not just bring as many toys as possible?”

Dorara thinks this is a brilliant idea! She grabs her backpack, which has a maximum weight capacity of K .

Luckily, Dorara is a very organized kid. She has already arranged all her N toys in a row, sorted from the lightest to the heaviest.

Your task is to help Dorara to pack her toys one by one, starting from the lightest, and tell her the maximum number of toys she can carry without exceeding the backpack’s limit.



Input Format

The input consists of multiple test cases until EOF. Each test case contains two lines. The first line contains two integers N and K , representing the total number of toys, and the weight capacity of the backpack. The second line contains N integers, $w_1, w_2, \dots, w_N$ representing the weights of these N toys.

Note: The weights are guaranteed to be sorted in ascending order (small to large).

Output Format

For each test case, output the maximum number of toys Dorara can bring.

Technical Specification

- $1 \leq N \leq 100$
- $1 \leq K \leq 10,000$
- $1 \leq w_i \leq 1,000, 1 \leq i \leq N$

Sample Input 1

```
5 10
1 2 3 4 5
3 5
```

Sample Output 1

```
4
0
```

6 7 8
